Coral Reefs (2016) 35:839–849
DOI 10.1007/s00338-016-1421-4

REPORT

Temporal consistency in background mortality of four dominant coral taxa along Australia's Great Barrier Reef

C. Pisapia[1,2] · K. D. Anderson[1] · M. S. Pratchett[1]

Received: 11 May 2015 / Accepted: 8 February 2016 / Published online: 5 April 2016
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Abstract Studies on the population and community dynamics of scleractinian corals typically focus on catastrophic mortality associated with acute disturbances (e.g., coral bleaching and outbreaks of crown-of-thorns starfish), though corals are subject to high levels of background mortality and injuries caused by routine and chronic processes. This study quantified prevalence (proportion of colonies with injuries) and severity (areal extent of injuries on individual colonies) of background mortality and injuries for four common coral taxa (massive *Porites*, encrusting *Montipora, Acropora hyacinthus* and branching *Pocillopora*) on the Great Barrier Reef, Australia. Sampling was conducted over three consecutive years during which there were no major acute disturbances. A total of 2276 adult colonies were surveyed across 27 sites, within nine reefs and three distinct latitudinal sectors. The prevalence of injuries was very high (>83%) across all four taxa, but highest for *Porites* (91%) and *Montipora* (85%). For these taxa (*Montipora* and *Pocillopora*), there was also significant temporal and spatial variation in prevalence of partial mortality. The severity of injuries ranged from 3% to more than 80% and varied among coral taxa, but was fairly constant spatially and temporally. This shows that some injuries have considerable longevity and that corals may invest relatively little in regenerating tissue over sites of previous injuries. Inter-colony variation in the severity of injury also had no apparent effect on the realized growth of individual colonies, suggesting that energy diverted to regeneration has a limited bearing on overall energetic allocation, or impacts on other life-history processes (e.g., reproduction) rather than growth. Establishing background levels of injury and regeneration is important for understanding energy investment and life-history consequences for reef-building corals as well as for predicting susceptibility to, and capacity to recover from, acute disturbances.

Keywords Disturbance · Resilience · Coral reefs · Partial mortality · Temporal variation · Long-term

Communicated by Ecology Editor Dr. Alastair Harborne.

✉ C. Pisapia
chiara.pisapia@my.jcu.edu.au

[1] ARC Centre of Excellence for Coral Reef Studies, James Cook University, Townsville, QLD 4811, Australia

[2] AIMS@JCU, Australian Institute of Marine Science, School of Marine Biology, James Cook University, Townsville, Australia

Introduction

Disturbance ecology is largely focused on the effects of infrequent and unpredictable, but often severe and large-scale, disturbances, such as fires, floods and outbreaks of pest species (Scatena and Lugo 1995; Lugo and Scatena 1996). These acute disturbances tend to cause high levels of instantaneous mortality, especially among key habitat-forming organisms (Connell 1997), and may have strong selective pressures on community structure and ecosystem function (White and Pickett 1985). It is important to realize, however, that these acute disturbances operate against a backdrop of more persistent or chronic disturbances that can exert significant influence on mortality schedules of colonies and influence the outcomes of biological interactions (Connell et al. 1997; De'ath et al. 2009). Ultimately, the incidence and severity of injuries may also affect a colony's susceptibility to, and recovery from, other acute disturbances (Andres and Witman 1995).

Coral reefs are renowned as highly dynamic ecosystems, strongly structured by acute disturbances (Connell 1997). On Australia's Great Barrier Reef (GBR), for example, sustained declines in live coral cover over the last 27 yr have been largely attributed to high-intensity storm events and episodic outbreaks of crown-of-thorns starfish (De'ath et al. 2012). It is likely, however, that background rates of coral mortality and injury have also increased over this period, owing to increased intensity of chronic disturbances and sustained environmental change (Wakeford et al. 2008). While background mortality rates (both partial and whole-colony mortality) of reef-building corals have been shown to be significant (e.g., Done 1988; Bythell et al. 1993; Madin et al. 2014; Pisapia and Pratchett 2014), few studies have specifically documented temporal dynamics in the background mortality and injury regimes of scleractinian corals.

In colonial organisms such as reef-building corals, mortality can occur at the level of the polyps, resulting in partial mortality or injury (Hughes and Connell 1987) or at the level of the colony (e.g., Madin et al. 2014). Rates of whole-colony mortality are a key component of the life-history and population ecology of different corals (Wakeford et al. 2008; Sweatman et al. 2011; De'ath et al. 2012). However, rates of partial mortality or injuries can also have an important influence on coral demography (Hughes and Jackson 1985; Baird and Marshall 2002). Injured colonies often allocate energy toward repairing and regenerating damaged tissues and skeleton, diverting energy from other metabolic processes, principally reproduction (Hall 1998). Corals may also be subject to extremely frequent and sometimes very severe injuries caused by predation (Cumming 2002; Rotjan and Lewis 2008; Lenihan et al. 2011; Williams and Miller 2012), accumulation of sediment (Guzman et al. 1994; Nugues and Roberts 2003), competition (Henry and Hart 2005) and/or pathogenesis (Williams and Miller 2012).

Once injured, corals may or may not repair damaged tissues and/or skeleton (Bak and Steward-Van Es 1980; Meesters et al. 1994). Since colonies often fail to regenerate large injuries (from 5 cm^2) (Bak and Steward-Van Es 1980; Meesters et al. 1994) with subsequent reduction in survival potential of an individual (Cumming 2002), partial mortality greater than 50% of the colony surface area was considered high. The capacity for regeneration and rates of tissue repair vary greatly among coral species (Liddle and Kay 1987; Liddle 1991; Meesters and Bak 1994; Meesters et al. 1997; Oren et al. 1997), mainly due to inherent differences in energy allocation, but they also depend on the local environment, disturbance history (Henry and Hart 2005), and the size and type of injury (Bak and Steward-Van Es 1980; Hall 1997). Small injuries generally cause localized declines in growth (Bak 1983; Guzman et al. 1994; Meesters et al. 1994; Lirman 2000) and reproduction (Hall 1998), whereby injured parts of the colony may temporarily stop growing. It is clear, however, that larger injuries can affect overall growth and reproduction of colonies (Rinkevich and Loya 1989; Meesters et al. 1994).

The purpose of this study was to resample individual coral colonies over a 3-yr period to provide explicit estimates of the rate of injuries and the fate of injured coral colonies. Sampling was intentionally spread across distinct latitudinal sectors, testing for large-scale variation in the prevalence and severity of injuries among common coral taxa, following Pisapia and Pratchett (2014). This study builds on initial results of Pisapia and Pratchett (2014), who reported instantaneous measures of partial mortality based on a single survey as per Hall (1997) and Lirman et al. (2013). The problem with instantaneous measures of coral injury is that taxonomic comparisons in rates of injury may be confounded by differences in the time taken for injured colonies to regenerate or die. Longer-term studies on individual colonies also provide significant additional insights on the persistence of injuries, the effects of background mortality on coral growth and the ultimate fate of coral colonies subject to extremely severe injuries.

Methods

Study sites

This study was conducted within each of three distinct locations separated by 4–5° of latitude along Australia's GBR: the northern sector in the vicinity of Lizard Island (14°41′S, 145°28′E); the central sector, in vicinity of Trunk Reef (18°25′S, 146°47′E); and the southern sector, in the vicinity of Heron Island (23°27′S, 155°55′E) (Fig. 1). Three mid-shelf reefs were sampled within each location; at each reef, three sites were sampled along the exposed reef crest. At each site, three replicate 10 × 5-m belt transects orientated parallel to and positioned within 5–10 m of the reef crest were marked using stainless steel stakes, following Trapon et al. (2013). Transects were established, and corals initially surveyed in 2011 (Pisapia and Pratchett 2014). All corals were then re-surveyed every 12 months from 2011 to 2014, to capture and compare annual mortality and injury regimes within and among common coral taxa.

Partial mortality

This study focused on four distinct coral taxa—*Acropora hyacinthus*, branching *Pocillopora*, massive *Porites* spp. and encrusting *Montipora* spp.—which were selected for their high abundance across all study sites (Trapon et al.

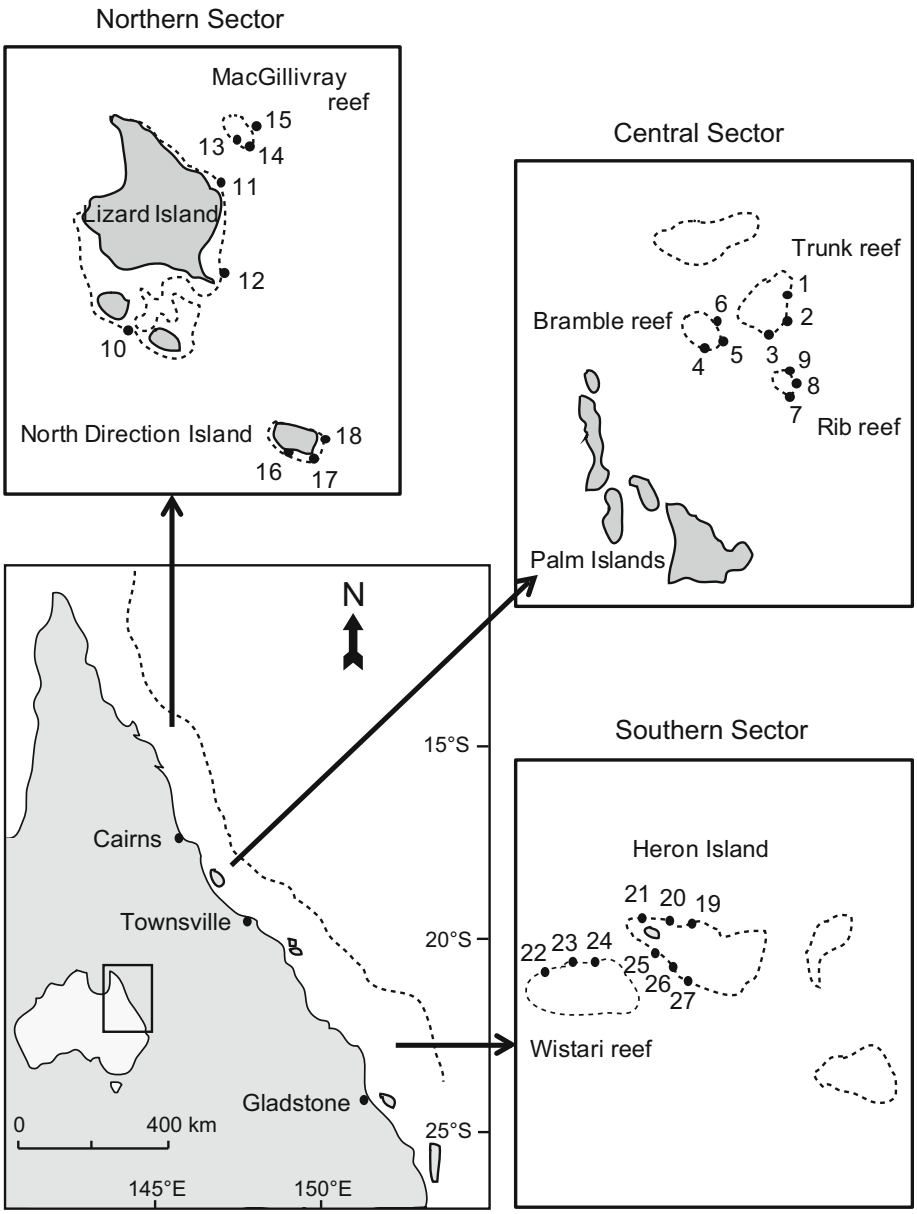

Fig. 1 Map of the three study sectors (northern, central and southern) along the Great Barrier Reef, Australia. Within each latitudinal sector, three mid-shelf reefs, each with three sites on the exposed reef crest (*numbers*), were sampled once per year from 2011 to 2014. Figure adapted from Trapon et al. (2013)

2013), but also because of their contrasting morphologies and life-history strategies (Darling et al. 2012). To quantify prevalence and severity of partial mortality, all living adult colonies (>10 cm) of each of the four taxa within each transect ($n = 200$–300 per sector) were surveyed every year. Each year, every colony was photographed from the top with appropriate scale reference. Photographs were then processed using the software Image J (http://imagej.nih.gov/ij), to precisely estimate the areal extent of live tissue of each colony. The prevalence and severity of injuries were, however, recorded in situ due to difficulties in capturing the full extent of injuries (especially on the periphery and base of colonies) based on a single photograph. The extent of injuries recorded for individual colonies ranged from zero (no apparent injuries) through to 100% (whole-colony mortality). It is possible that in colonies that can fragment (e.g., branching *Pocillopora* and plating *A. hyacinthus*) partial mortality was under-estimated.

Data analyses

Taxonomic differences and temporal and spatial variation in prevalence of partial mortality (specifically, the proportion of injured colonies per transect) were analyzed using generalized linear mixed models (GLMMs) with a binomial link-function (e.g., logistic regression) (Warton and Hui 2011), comparing among taxa, years, colony size, sectors and reefs nested within sectors. Similarly,

taxonomic differences and temporal and spatial variation in severity of partial mortality (areal extent of injuries across individual colonies) were analyzed using a GLMM with a binomial link-function, using a hierarchically nested design, with transects (three per site) nested within site, sites (three per reef) nested within reef and reefs (three per sector) nested within sectors (three sectors). Colony size and taxon were also treated as independent variables to investigate whether partial mortality was interactively affected by coral taxon and colony size. More specifically, the fixed factors were sector taxon, colony size and years, while reef, site and transect were random factors. To triangulate the spatial scale that captured most variation in partial mortality, models with different nested structures were compared by maximum likelihood and the model with the lowest Akaike information criterion (AIC) was chosen as the best representation of the variation in the data.

To test whether the severity of partial mortality on individual colonies impacted their subsequent growth rates, we explored the relationship between changes in the areal extent of live tissue for each individual coral colony from year 1 to year 3 and time-averaged partial mortality (colonies with partial mortality >0%) recorded for each colony. The significance of these relationships was tested using a GLMM for each coral taxon with growth as the response variable and mortality as a continuous predictor. Reef (three per sector), site (three per reef) and transect (three per site) were random factors, and data were aggregated over the full duration of the study.

Results

Prevalence of partial mortality

A total of 2276 colonies were surveyed along the GBR to assess background rates of injury and mortality, including 862 colonies of *A. hyacinthus*, 301 colonies of massive *Porites*, 505 colonies of branching *Pocillopora* and 608 colonies of encrusting *Montipora*. Whole-colony mortality was between 1 and 8% per annum. More specifically, 52 (of 862) *A. hyacinthus*, 17 (of 608) *Montipora* encrusting, 38 (of 505) branching *Pocillopora* and two (of 301) *Porites* spp. experienced whole-colony mortality.

Prevalence of partial mortality (the proportion of colonies with visible injuries) was very high across all taxa, averaging 71% for *Acropora*, 59% for branching *Pocillopora*, 85% for encrusting *Montipora* and 92% for *Porite*s, across all sectors, reefs and sites (Fig. 2). Coral taxa showed significant differences in the distribution of degrees of partial mortality (Table 2). *Porites* and *Montipora* had slightly higher proportion of colonies with partial mortality >50% of the colony area compared to *A. hyacinthus* and *Pocillopora* (Table 2; Fig. 3). However, frequency of partial mortality was high in all coral taxa; (>25% of colonies showed degrees of partial mortality ranging between 50 and 80% of surface area) (Table 2; Fig. 3).

When analyzed in each taxon separately, the proportion of colonies with partial mortality >50% of the colony area was high and varied slightly among latitudinal sectors pooled across 3 yr (Fig. 4); prevalence of partial mortality was higher in the southern than in the central and northern GBR for all coral taxa (Fig. 4).

All the injuries were covered in algae, and some injuries were relatively old with corallite structures gone or covered by algae and/or other organisms. White exposed skeleton indicating recent injury (a few days prior to the survey) were noticed only as grazing marks on some massive *Porites*. Over the 3 yr some injuries shrank indicating signs of regeneration, but the size of most injuries did not vary with time. Interestingly, diseases were only recorded on a few *A. hyacinthus* colonies in the central GBR in the first year and three affected colonies died during the study period.

Prevalence of partial mortality was consistently high across all years, with no significant variation among years (variance 1%; Table 1), except for *Pocillopora* spp., which showed a slight increase in the proportion of new colonies with injuries in the third year, and only in the southern sector (Fig. 4). The prevalence of injury was also consistent among sectors and reefs; most variation occurred among sites within reefs rather than among sectors or reefs (Table 1). Prevalence of partial mortality was significantly affected by taxon (variation 10%; Table 1), with *Montipora* and *Porites* showing significantly higher prevalence of partial mortality than *Pocillopora* and *A. hyacinthus* (Table 1; Fig. 4). Conversely, colony size did not have an effect on the proportion of injured colonies (Table 1).

Few new injuries were apparent through the course of the study (Fig. 5). However, in all coral taxa the proportion of new colonies injured in the second and third year slightly increased in most sectors (Fig. 5). In *A. hyacinthus*, massive *Porites* and branching *Pocillopora*, the proportion of new colonies injured in the third year was higher than colonies injured in the second year (Fig. 5), while the proportion of colonies that died remained low over the study period (<20%; Fig. 5).

Severity of partial mortality varied mainly among reefs (Table 2); however, the greatest variation was observed within transects (96.6%; Table 2). Colony size did not have any effect on severity of injury, while coral taxon affected the extent of partial mortality (Table 2). However, the

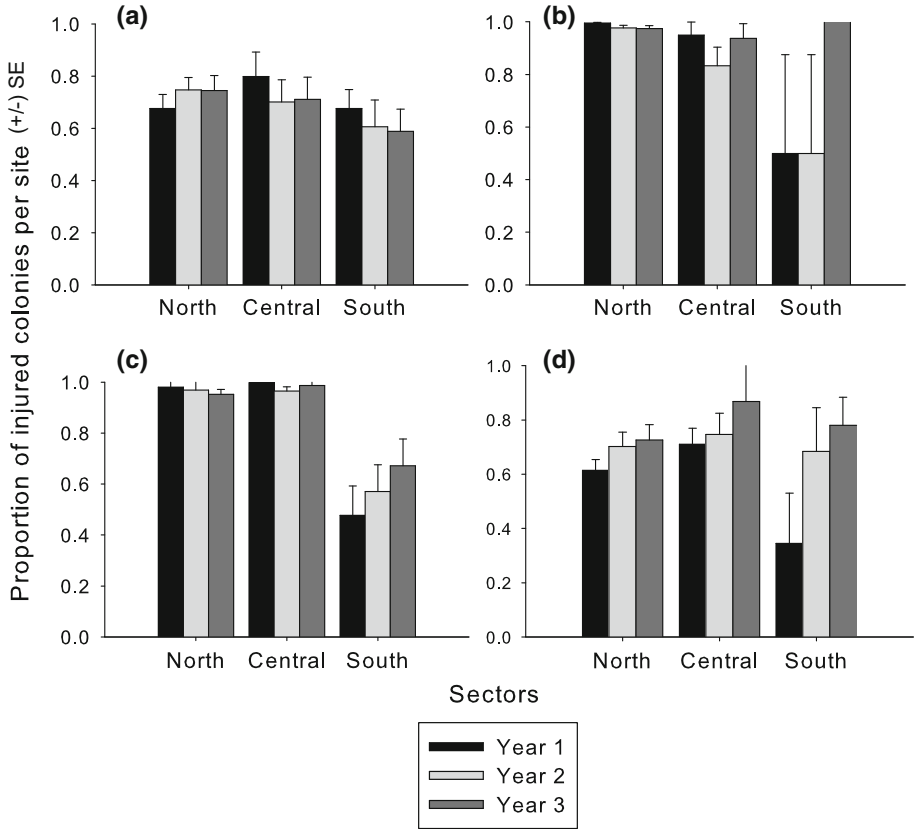

Fig. 2 Prevalence of mortality (mean proportion of injured colonies) per site in 3 yr in three latitudinal sectors (north, central and south) in four coral taxa. **a** *Acropora hyacinthus*, **b** massive *Porites*, **c** encrusting *Montipora* and **d** branching *Pocillopora*. *Error bars* are standard errors

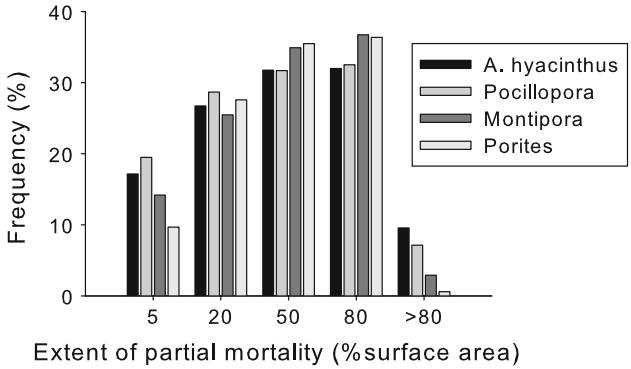

Fig. 3 Frequency distribution of the extent of partial mortality (percent surface area loss) for four taxa (*Acropora hyacinthus*, *black bars*; branching *Pocillopora*, *mid-gray bars*; encrusting *Montipora*, *dark gray bars*; massive Porites, *light gray bars*) pooled across 3 yr

variation explained by coral taxon was still very low (3.4%; Table 2). The severity of injury of encrusting *Montipora* was significantly lower (9.0%) in the northern sector compared to the other two sectors (24 and 16%) and rose to 20% over the study period (Fig. 4). Severity of partial mortality varied through time only in *Pocillopora* (Fig. 4), but there was an increase in severity of injury of *Porites* from 5.4 to 20% from the second year (2012–2013) to the third (2013–2014) in the southern sector (Fig. 4).

Effects of partial mortality on colony growth

Although severity of partial mortality ranged from 2 to 80% among colonies, there was no obvious effect of the proportional extent of injuries (measured to the nearest 5% in each year) on colony growth (Table 3). In all coral taxa, severe injury did not constrain the change in size of individual colonies at any latitude or any spatial scale.

Discussion

This study documented high prevalence and severity of partial mortality in four common coral taxa on Australia's GBR. While rates of whole-colony mortality were fairly moderate (between 1 and 8% per annum in the four coral taxa), virtually all colonies had recent and sometimes very severe injuries. The prevalence of injuries was particularly high for *Porites* and *Montipora* (91 and 85%, respectively), which is much higher than rates of injury reported elsewhere. In the Indian Ocean, for example, 27.5% of colonies of *Acropora cytherea* had recent injuries (Pratchett et al. 2013), while in the Florida Keys prevalence of injuries was reported to be generally <5% for *Agaricia* spp., *Diploria strigosa* and *Montastrea cavernosa* (Kramer 2003; Lirman et al. 2013). It is important to

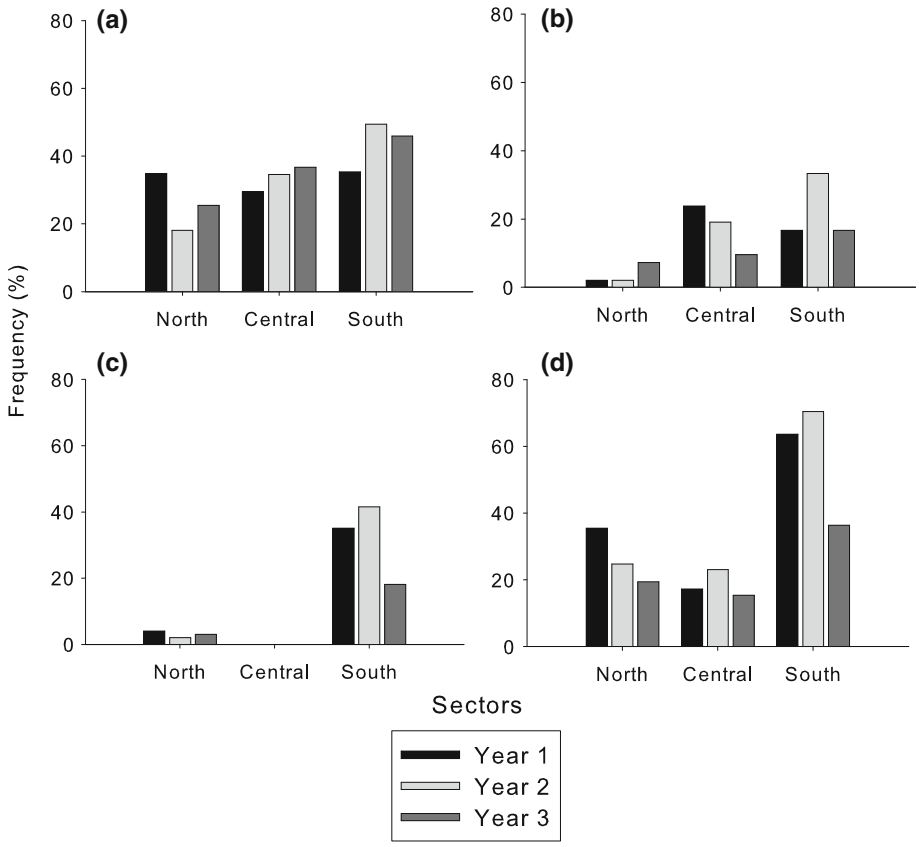

Fig. 4 Frequency distribution of severity of partial mortality (>50% surface area loss) in **a** *Acropora hyacinthus*, **b** massive *Porites*, **c** encrusting *Montipora* and **d** branching *Pocillopora* in three latitudinal sectors, the northern, central and southern Great Barrier Reef from 2011 to 2014. The data were pooled across sites and reefs. Year 1 (*black bars*) refers to 2011–2012, Year 2 (*light gray bars*) refers to 2012–2013 and Year 3 (*dark gray bars*) refers to 2013–2014

Table 1 Results of a generalized linear mixed model with a binomial link-function to test for temporal and spatial variation in prevalence of injury (proportion of injured colonies per site) from 2011 to 2014

	df	Wald statistic	*p*	% Variation explained
Sector	4	0.38	0.53	8.8
Reef (sector)	2	0.09	0.92	21
Year	4	0.03	0.98	1
Taxon	4	0.02	0.05	10
Size	8	0.04	0.88	
Error	61			59.2

Percentage of variation associated with particular spatial scales is also included. Model building results: lowest AIC = 939.7, *L* ratio $\chi^2 = 1023.1$, $p = 0.000$

note, however, that each of these studies defined injuries as areas of recently exposed skeleton yet to be colonized by turf algae (Lirman et al. 2013; Pratchett et al. 2013) and explicitly ignored older injuries whose origins it is impossible to date. In this study, we aggregated both recent and established injuries, resulting in much higher estimates of both prevalence and severity of injuries. Even though corals are more likely to invest in tissue repair for recent and smaller injuries (Hall 1997), our justification for measuring the total area of both recent and established injuries on every colony is that in any reductions in live tissue area will constrain the capacity of corals to acquire energy for maintenance, growth or reproduction (Hughes and Jackson 1985).

The few studies that have documented natural rates of injuries (Hughes and Jackson 1985; Hall 1998; Cumming 2002; Bruckner 2012; Lirman et al. 2013; Pisapia and Pratchett 2014) have reported different extents of partial mortality ranging from 5% (Hall 1998; Cumming 2002) to <30% (this study and Pisapia and Pratchett 2014) up to 99% (Bruckner 2012). However, these studies only provided static measures involving a single assessment, often at small spatial scales. Pisapia and Pratchett (2014) did investigate partial mortality over a large spatial scale, but they only recorded instantaneous measures of injury missing any temporal variation and taxonomic differences in partial mortality, unlike this study. Even though it is critical to investigate the natural occurrence of tissue loss, the key parameter that has not been investigated yet and will ultimately determine recovery capacity of corals, is the change in rates of tissue loss. This study explicitly tested for changes in injury rates through time, following individual corals over 3 yr, but documented limited change in rates of tissue loss as both temporal and spatial variation in prevalence and extent of injury was minimal. This suggests that background mortality is relatively constant in space and time at the population level. It is possible that most

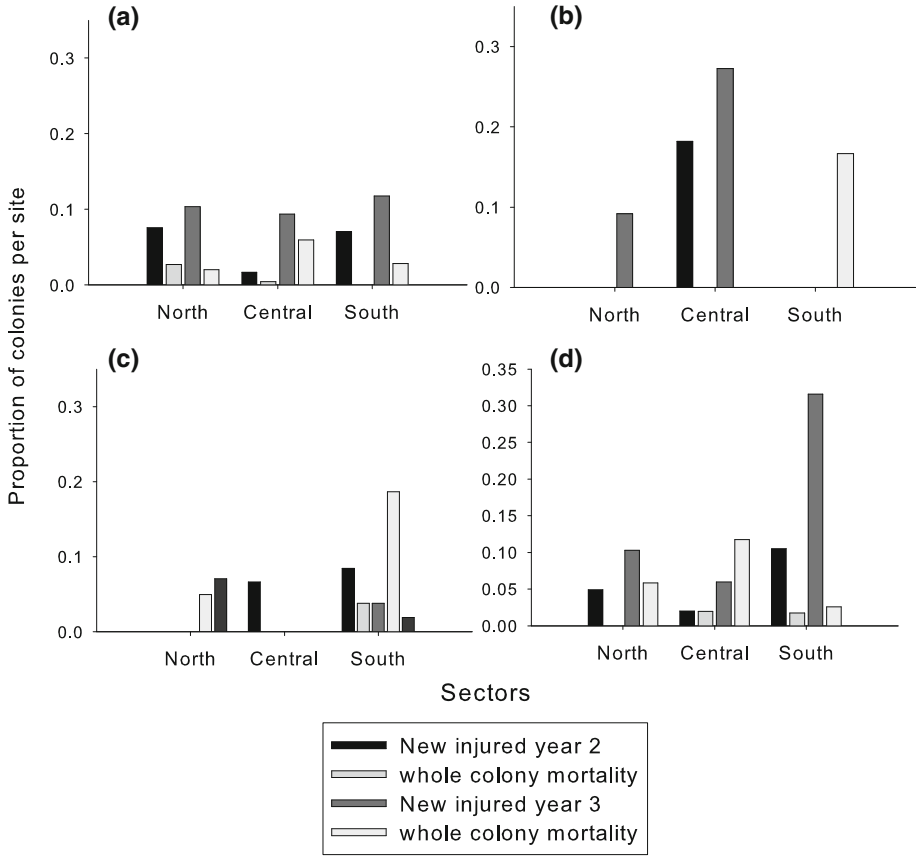

Fig. 5 Proportion of new colonies injured and whole-colony mortality in Year 2 (2012–2013) and Year 3 (2013–2014) for **a** *Acropora hyacinthus*, **b** massive *Porites*, **c** encrusting *Montipora* and **d** branching *Pocillopora* in three latitudinal sectors, the northern, central and southern Great Barrier Reef

Table 2 Results of a generalized linear mixed model with a binomial link-function to test for temporal and spatial variation in severity of injury (specifically, the proportion of dead versus living components of each colony) for four coral taxa from 2011 to 2014 at all spatial scales

	df	Wald statistic	*p*	% Variation explained
Sector	2	0.18	0.14	
Reef (sector)	4	9.49	<0.001	
Site (reef)	4	7.56	0.06	
Transect (site)	2	5.31	0.14	
Year	3	8.31	0.05	
Taxon	2	29.1	<0.001	3.4
Size	2	9.54	0.22	
Error	73			96.6

Model building results: lowest AIC = 235.0, *L* ratio $\chi^2 = 598$, $p = 0.000$

studies may overestimate the commitment of corals to repair and regeneration, such that nearly all corals have a legacy of injuries that accumulates through time. This may in turn explain why there was limited spatial and temporal variation at the scales in this study.

Although we only compared among four taxonomic groups, there is evidence that taxonomic differences in prevalence of partial mortality are related to morphology. Massive and encrusting corals showed higher prevalence (>80%) and frequency of partial mortality compared to plate and branching corals. Previous studies have suggested that branching corals are disproportionately susceptible to many routine agents of coral injuries (e.g., breakage: Meesters et al. 1996; predation: Cole and Pratchett 2011; competition: Lang 1973), but taxonomic differences in prevalence of injuries probably relate more to differences in rates of somatic growth and repair (Meesters et al. 1992, 1996, 1997; Hall 1997) than susceptibility to disturbances per se. Notably, branching corals (e.g., *Acropora* and *Pocillopora*) have relatively high regeneration capacity, rapid linear growth, high rates of recruitment, high rates of mortality and short generation time (less than 30 yr), all resulting in highly dynamic populations with rapid turnover (Yap et al. 1992; Adjeroud et al. 2007). Conversely,

Table 3 Results of a generalized linear mixed model for each coral taxon to test whether severity of partial mortality affected colony growth

	df	Wald statistic	*p*	AIC	*L* ratio χ^2	*p*
Acropora hyacinthus				7135.8	0.04	0.82
Reef (sector)	8	9.32	0.54			
Site (reef)	4	1.34	0.86			
Transect (site)	3	0.74	0.68			
Partial mortality	4	0.02	0.53			
Error	61					
Porites				1705.4	0.86	0.35
Reef (sector)	3	4.82	0.27			
Site (reef)	3	3.21	0.34			
Transect (site)	3	0.14	0.92			
Partial mortality	3	0.35	0.55			
Error	80					
Montipora				2819.2	0.004	0.94
Reef (sector)	3	0.01	0.89			
Site (reef)	4	2.1	0.68			
Transect (site)	4	1.43	0.92			
Partial mortality	4	0	0.48			
Error	65					
Pocillopora				3168.0	9.25	0.024
Reef (sector)	4	5.9	0.5			
Site (reef)	4	5.31	0.15			
Transect (site)		4.79	0.09			
Partial mortality	3	4.14	0.41			
Error	73					

Growth was the response variable and mortality the continuous predictor. Reef (three per sector), site (three per reef) and transect (three per site) were random factors. Data were aggregated over the 3 yr of the study

massive *Porites* invest proportionally less energy in regeneration, which is generally slow (Denis et al. 2011), and have slow growth rates, large colony size, low rates of whole-colony mortality, low rates of recruitment and long life span, all resulting in slow population turnover (Bak and Engel 1979; Adjeroud et al. 2007). Consequently, injuries apparent on large, slow-growing corals are likely to have accumulated over a much longer period (Meesters et al. 1996; Denis et al. 2011) resulting in higher prevalence and severity of injury.

While this is the first study to explicitly quantify variation in the prevalence of background mortality along the length of the GBR, previously published data documented marked latitudinal differences in population replenishment (highest in the northern GBR) despite similar adult abundances (Hughes et al. 1999), suggesting that there may be marked differences in the underlying dynamics of coral populations along the GBR. Based on observed patterns, it would be expected that background rates of mortality would be highest in the northern GBR and lowest in southern GBR. However, this study documented limited variation in prevalence or severity of partial mortality along the GBR. Large-scale trends were only observed in *Montipora* with lower prevalence of partial mortality in the southern sector. However, there was greater variation in the severity of injuries among adjacent colonies within the same reef habitat than among colonies in different latitudinal sectors. This shows that the lack of conformity between recruitment and adult abundance highlighted by Hughes et al. (1999) is likely due to differences in growth and mortality of newly settled corals or latitudinal variation in adult growth rather than differences in rates of tissue loss.

Specific drivers of the observed small-scale (reef, site and transect) differences in the prevalence and severity of coral injuries remain unclear, as the source of mortality was rarely apparent. Even though it is critical to identify the direct links between the stress and response, this is almost impossible in complex ecosystems like coral reefs (Lirman et al. 2013). In corals, partial mortality is a general stress response and to attribute the cause of injury to a specific agent, the sampling frequency and intensity needs to be very high and integrated with controlled laboratory studies, and/or the disturbance event must be clearly

identifiable (such grazing parrotfish bites). However, some colonies, especially *A. hyacinthus* and massive *Porites*, also showed injuries completely surrounded by living tissue. This sort of tissue damage is harder to identify, but may be due to senescence (Rinkevich and Loya 1989), fish bites, past incidence of disease, sedimentation and/or direct damage, such as that caused by divers or anchors (Meesters et al. 1996).

High prevalence and severity of injuries will almost certainly constrain coral growth. Following injury, resources are often redirected into regeneration, and this change in energy allocation may limit resources available for other demographic processes, such as growth (Meesters et al. 1994. The patterns of resource allocation will depend upon the degree of physiological integration (Bazzaz et al. 1987). Growth is often reduced in regenerating corals (Bak 1983; Liddle and Kay 1987; Guzman et al. 1994; Meesters et al. 1994; Lirman 2000). However, sometimes growth rates may be faster in injured colonies than in uninjured ones (Loya 1976). In this study, prevalence and severity of partial mortality had no bearing on the realized growth of colonies of any coral taxon. This is an important finding and tends to confirm the findings of previous studies (e.g., Hall 1998) that suggested that the energy allocated to regeneration may be diverted from reproduction, rather than growth. It is also possible that energy diverted to tissue repair may have negligible impact on the overall energy budget, especially if tissue repair imposes a smaller energetic cost on the colony than investment in new skeletal material (Hall 2001).

Consistently high background levels of prevalence of coral injury (e.g., more than 50% of corals) have the capacity to significantly affect fitness of individual colonies and hence reduce the capacity of populations to recover between major acute disturbances (Johnson and Preece 1992; Langmead and Sheppard 2004; Habeeb et al. 2007; Wakeford et al. 2008; Bruckner and Hill 2009; Fung et al. 2011). Recovery rates following bleaching are likely to be lower in injured colonies than in uninjured ones (Meesters and Bak 1994 Baird and Marshall 2002; Fine et al. 2002). However, recovery rates largely depend on current levels of lipid reserves (Anthony et al. 2009). High levels of lipids in coral tissues is vital for survival (Bachok et al. 2006), because corals tend to consume their lipid reserves when maintenance costs exceed carbon acquisition such as during a bleaching event (Anthony et al. 2009).

Prevalence and severity of background mortality have significant ramifications for corals' capacity to cope with increasing acute disturbances, such as climate-induced coral bleaching. Colonies with low partial mortality and hence in good physiological condition may have a greater potential to withstand and recover from environmental changes (i.e., are more resilient) than conspecifics with high partial mortality and hence in poor condition, and will therefore have a disproportionate role in population recovery. Background mortality rates have rapidly increased in temperate forests (van Mantgem et al. 2009), and if background whole-colony mortality and partial mortality are also increasing on coral reefs, the capacity for recovery of these ecosystems will likely be dramatically reduced.

Acknowledgments This study was funded by the ARC Centre of Excellence of Coral Reef Studies, AIMS@JCU, a Graduate Research Scheme award and a Great Barrier Reef Marine Park Authority Science for Management award. The authors are indebted to V. Messmer for comments on the manuscript and to M. Trapon, J. Casey, A. Hoey, D. Coker, D. Schmid-Lieberg, S. Katz for assistance in the field, the staff at Lizard Island, Orpheus Island and Heron Island Research Stations, and the *RV Kirby* for field and logistical support.

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